



Context

You are a team of engineers at AeroTagus, a Portuguese company specializing in the design and manufacture of Unmanned Aerial Vehicles (UAVs). You have been tasked with developing a new aerial surveillance drone for a defense customer.

The platform, internally designated Vigilante, was designed for long-endurance Intelligence, Surveillance, and Reconnaissance (ISR) missions. These are the minimum requirements defined by the client:

- **Fixed wing with electric propulsion**
- **Endurance:** $\geq 2:40$ h with a camera turret as payload
- **Minimum Cruise speed:** ~ 70 km/h
- **Envelope:** 3000 m above ground level
- **Max dimensions:** 3.5m wingspan, 2m of length and 20kg empty weight

Team

You are a team composed of 4 specialists with distinct responsibilities:

- **Aerodynamics Specialist:** Responsible for the UAV aerodynamic design and thermal management of electrical propulsion system, optimizing its performance (lift, drag, stability, cooling, endurance).
- **Structures Specialist:** Responsible for ensuring the structural integrity of the UAV and in minimizing its weight.
- **Propulsion Specialist:** Responsible for managing the design, integration and troubleshooting of the UAV's electrical propulsion components.
- **Test Specialist:** Responsible for flight test campaigns, analysing general performance from test data and providing feedback for improvement.

Task

The client defined a minimum endurance (with payload) of 2:40h. However, during flight testing, the team realized that requirement hadn't been met. You have a maximum of 45 minutes to complete the following:

1. Identify a **single, most likely root cause** of the endurance shortfall.
2. The next flight testing is scheduled to happen in one month. Propose a **specific, actionable solution**, that could be implemented in the UAV in the **timeframe of a month** before that test.

When you finish, state very clearly what is your answer to both these questions.

You must communicate **only by chat messages**. You must not share anything besides text messages.

You mustn't copy paste anything directly from your handout.

You must not use any other external tool, like ChatGPT.

All communication must be made in English.

You can transcribe voice into text so you can send messages more quickly using Speechnotes: <https://speechnotes.co/dictate/>

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UAV Initial Design

This section features the initial design decisions. Some small aspects of the design may have suffered small changes after bench testing.

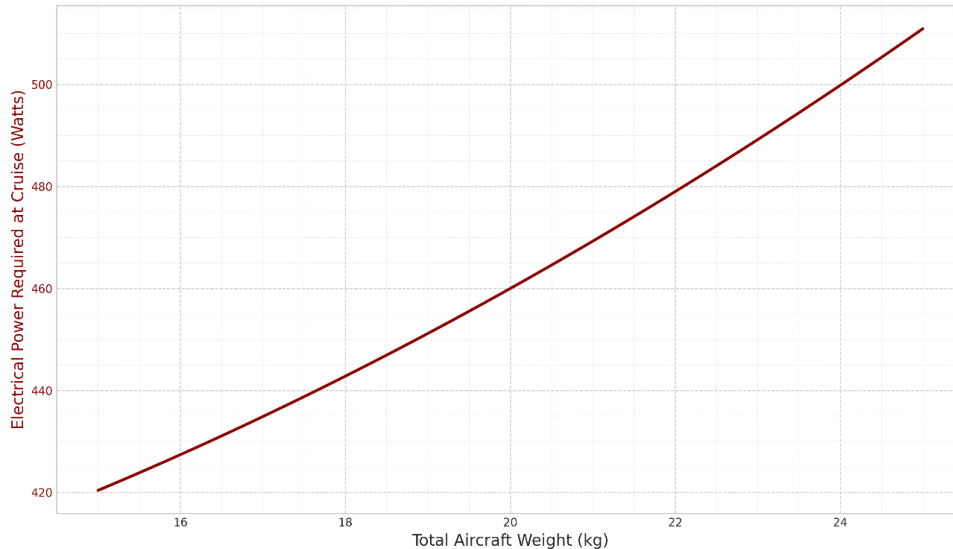
Structural Design and Materials

- **Primary Structure:** The fuselage, wings, and tail assembly are constructed from Carbon Fiber Reinforced Polymer (CFRP) composites.
- **High-Stress Components:** Most critical moving parts, such as control surface hinges and the payload gimbal, are machined from Aluminum 7075-T6.
- **Propeller:** 3D printed high-strength ABS, allowing for replacements to be rapidly executed on the field.
- **Battery Pack casing:** Polyamide with rubber supports to protect from vibrations.

Material	Key Use	Density (g/cm ³)	Tensile Strength (MPa)	Elastic Modulus (GPa)	Thermal Conductivity (W/m·K)
CFRP Composite	Fuselage, Wings, Tail	1.56	1500	150	5
Aluminum 7075 T6	Moving parts, Gimbal	2.81	510	71	130
3D Printed ABS	Propeller, Elec. Bay	1.13	60	3	0.25
Polyamide	Battery Pack Casing	1.10	75	2.5	0.2
Silicone Foam	Motor termal isolation	0.25	0.60	0.05	0.06

Performance vs Weight

UAV Power vs. Weight at 80 km/h Cruise



Aerodynamic Design

- **Overall Configuration:** A conventional fixed-wing configuration was chosen, featuring a high aspect ratio wing to minimize induced drag, combined with a V-tail configuration and rear positioned propeller.
- **Fuselage Design:** The fuselage has a streamlined, low-frontal-area profile. The integration of the underside camera turret introduced a necessary aerodynamic disruption.
- **Landing Gear:** A fixed, aerodynamically faired tricycle landing gear was chosen for simplicity, reliability and weight reduction. It's unclear how much this will increase drag.
- **Thermal Management:** to cool down the battery pack and small electronic components, a small duct was designed, which can also be covered to protect the battery from damage if cold weather is expected.

Propulsion Components Selection

Motor: A **Brushless DC (BLDC) Outrunner motor** was chosen. Its outrunner configuration provides the high torque needed to efficiently spin a large-diameter propeller at lower RPMs, ideal for endurance.

Electronic Speed Controller (ESC): A high-performance ESC manages the motor, providing critical telemetry data, including voltage, current, and temperature.

Battery Pack: A Lithium Polymer (LiPo) battery with a 40,000 mAh capacity was designed to meet the endurance target with a 30min surplus.

Design Validation & Pre-Flight Testing

Following the initial design phase, a comprehensive testing campaign was executed to validate predictions and performance of different components and identify any potential design flaws prior to the first full-scale flight.

Structural Bench Testing

Static and Dynamic Load Testing

The airframe was mounted in a test rig and subjected to simulated aerodynamic forces up to its design limit loads. Key components were subjected to vibration and fatigue tests.

There were several modifications:

- **Fuselage Crack:** A 3 cm crack appeared on the lower fuselage at 95% of the limit load. A structural reinforcement weighting 300g was added to the aircraft. It's yet to be confirmed that impact this had on the endurance.
- **Propeller Fatigue:** The 3D-printed propeller showed fatigue cracks after approximately 20 hours of simulated use. Due to this poor durability, it was replaced with a more robust, but more expensive, carbon fibre version.
- **Battery Cooling Ducts:** The original complex battery cooling ducts were redesigned to simplify manufacturing, with the impact on cooling performance remaining to be assessed.
- **Motor Thermal Isolation:** The motor's possible high operating temperature posed a thermal risk to the CFRP fuselage. A layer of silicone foam insulation was added between the motor and the airframe to mitigate this.

Propulsion System Bench Testing

During the electric propulsion system testing, it was noted that the carbon fiber propeller produced about 3% less static thrust than a 3D printed version at the same power. At the predicted cruise thrust setting, the system drew approximately 470W at 6000 rpm, resulting in an estimated endurance of 3 hours and 10 minutes.

Wind Tunnel Testing

An aerodynamics campaign using a scale model validated the CFD predictions, confirming the aircraft's handling was within expectations, though the landing gear showed more flow disturbance than simulated.

Separately, the original cooling system was tested with the full-scale propulsion system at 30°C. While the motor performed as expected, a thermal issue was identified: during a 1-hour simulated cruise, the battery's temperature rose to 45°C, 25% higher than predicted.

Flight Testing

The team did a battery worth of testing, in the following conditions:

- Ambient temperature at ground level: ~ 20 °C
- Wind: Up to 5m/s, no precipitation and no clouds.
- Payload: electro-Optical/Infrared (EO/IR) camera turret weighting 1.85kg.

Parameter	Design Target	Measured
Empty weight	17.96 Kg	18.45 Kg
Gross weight	-	20.30 Kg
Length	1.85m	1.85m
Wingspan	3.42m	3.43m

A four-run flight test was conducted to assess performance:

- **Run 1 (Handling):** The UAV was stable and responsive. A minor left yaw was noted, traced to a wing defect, and corrected.
- **Run 2 (Max Speed):** The aircraft achieved its predicted top speed. However, the pilot noted the throttle felt "sluggish" on the final pass, and telemetry showed battery temperatures were slow to recover.
- **Run 3 (Climb/Descent):** Initial climb performance was good. However, climbs performed after the endurance cruise felt "less energetic" and required more throttle.
- **Run 4 (Endurance Cruise):** After 50 minutes, the pilot reported the UAV felt "heavier" and required progressively more throttle to maintain altitude. This aligned with telemetry data showing a sustained rise in component temperatures and a clear, progressive loss of system efficiency.

Comparison of Predicted vs. Observed Performance Parameters:

Parameter	Predicted (Design Target)	Observed (Flight Test)
Maximum Speed	44 m/s	42.4 m/s
Electrical Power input at maximum speed	3000 W	2948W
Current at maximum speed	80A	79A
Maximum	15000 rpm	14970 rpm
Maximum climbing rate	9.4 m/s	9.2 m/s
Motor rpm at cruise	6000 rpm	6120 rpm
Endurance	3h 10min	~2h 20min

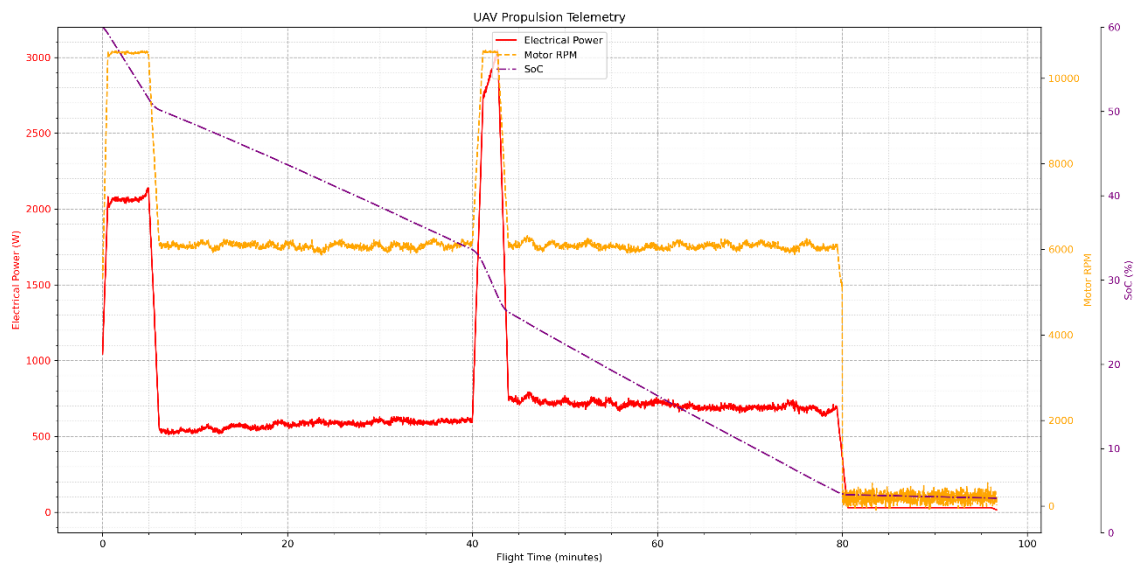
Telemetry from Run 4: Endurance Cruise

Below, there are some telemetry graphs from Run 4: Endurance Cruise. This run lasted for around 95min from taking off until landing, using all the remaining battery from the previous flights. The data gathered was used to estimate the total autonomy.

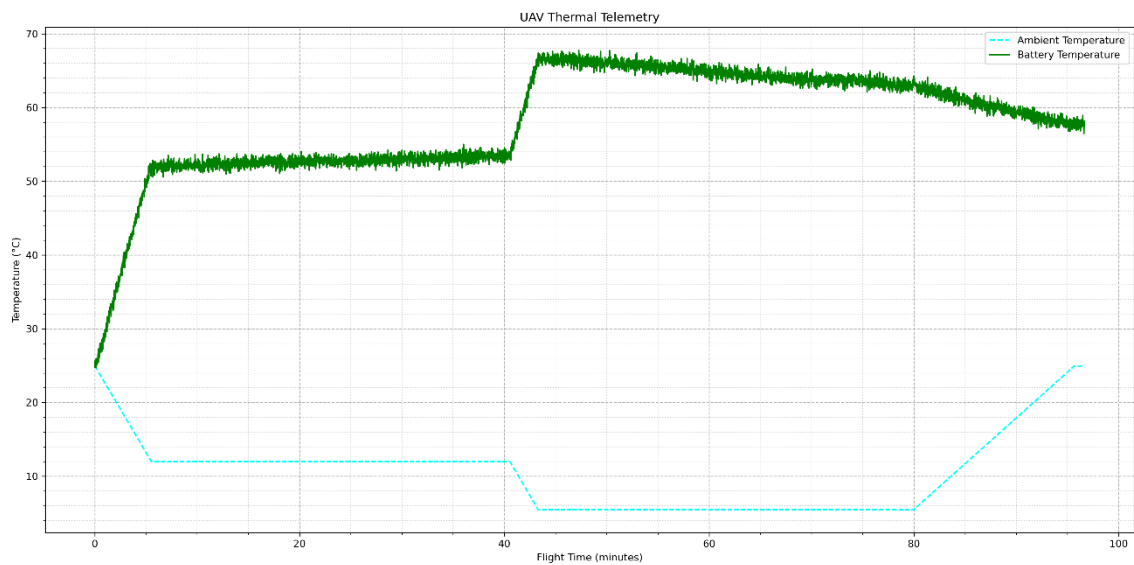
UAV Speed & Altitude Profile over Flight Time



Motor RPM, Power and Battery SOC over Flight Time



Ambient Temperature & Battery Temperature over Flight Time



Glossary

Based on the provided document, here are some of the key terms specific to aerospace engineering, categorized by discipline.

General Aircraft Concepts

Payload: Any equipment, cargo, or personnel carried by an aircraft that is not part of its basic structure, systems, or crew required for flight.

Envelope: The set of operational limits within which an aircraft can be safely flown. This typically includes boundaries for speed, altitude, load factor (g-forces), and manoeuvrability.

Airframe: The mechanical structure of an aircraft, including the fuselage, wings, empennage (tail), and landing gear.

Gimbal: A pivoted support mechanism that allows an object (like a camera, sensor, or engine) to remain stable or rotate independently about one or more axes, compensating for the motion of the vehicle it is mounted on.

V-tail: An aircraft tail configuration (empennage) that merges the functions of the vertical and horizontal stabilizers into two surfaces set at a dihedral angle, forming a "V" shape.

Yaw: The rotational motion of an aircraft about its vertical (or normal) axis. This is the "side-to-side" movement of the aircraft's nose.

True Airspeed (TAS): The speed of the aircraft relative to the air mass through which it is flying, which might be different relative to the ground speed when there is wind.

Ground Speed: The horizontal speed of an aircraft relative to the surface of the Earth. It is the combination of the aircraft's true airspeed and the effect of wind (headwind or tailwind).

Aerodynamic Concepts

Airfoil: The specific cross-sectional shape of a wing. Its geometry is designed to generate an aerodynamic force—primarily lift—when moving through a fluid.

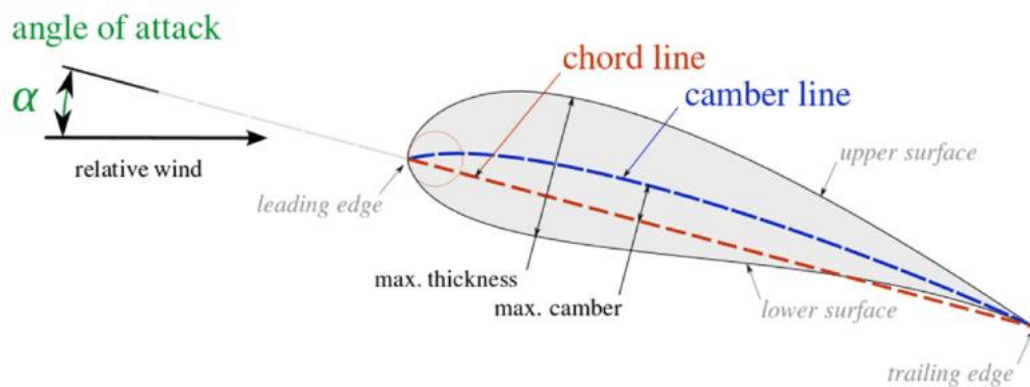
Chord: The straight-line distance between the leading edge (front) and the trailing edge (back) of an airfoil.

Mean Aerodynamic Chord (MAC): The average chord length of a tapered or non-rectangular wing. It is a specific chord value that, for aerodynamic purposes, represents the behavior of the entire wing.

Angle of Attack (α): The angle between an airfoil's chord line and the direction of the oncoming, undisturbed airflow.

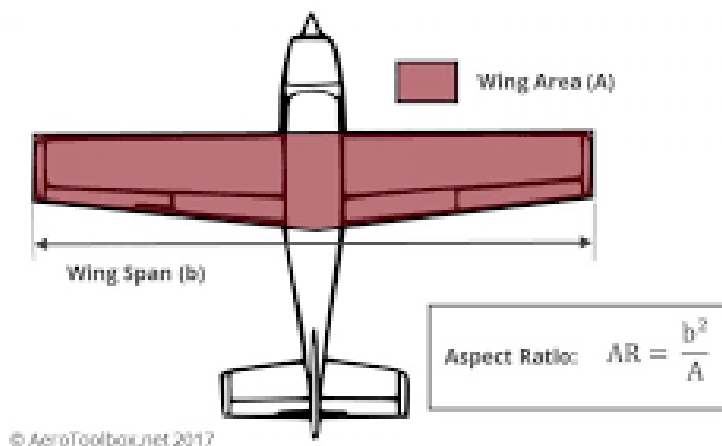
Lift: The component of the aerodynamic force acting on a body (like a wing) that is perpendicular to the direction of the oncoming airflow (the relative wind). It is the force that primarily opposes weight and allows an aircraft to fly.

Drag: The component of the aerodynamic force acting on a body that is parallel to and in the opposite direction of the oncoming airflow (the relative wind). It is the force that resists the motion of the aircraft through the air.



Wingspan: The maximum distance from one wingtip of an aircraft to the other.

Aspect Ratio: A dimensionless ratio describing a wing's shape, calculated as the square of the wingspan divided by the wing's area. High-aspect-ratio wings are long and narrow (like a glider's), while low-aspect-ratio wings are short and stubby (like a fighter jet's).



Reynolds number (Re): A dimensionless quantity used in fluid dynamics to predict airflow patterns. It represents the ratio of inertial forces to viscous forces and is critical for determining whether airflow will be laminar or turbulent, which significantly affects drag and lift.

Laminar Flow: A smooth, streamlined, and highly ordered fluid flow over a surface, characterized by fluid layers sliding past each other with little to no mixing. It typically results in very low skin friction drag.

Drag Polar: A graph that plots the drag coefficient (C_D) versus the lift coefficient (C_L) for an aircraft or airfoil. It is a fundamental tool for analyzing aerodynamic efficiency and performance (like range and endurance).

Stall: An aerodynamic condition where an increase in the angle of attack beyond a critical point causes the airflow to separate from the upper surface of the wing, resulting in a sudden and significant loss of lift and increase in drag.

Induced Drag: A component of aerodynamic drag that is an unavoidable byproduct of generating lift.

Structures Concepts

Structural Fatigue: The weakening or failure of a material caused by repeatedly applied loads or stress cycles. This damage can accumulate over time and lead to a fracture, even if the stress is below the material's ultimate tensile strength.

Tensile Strength: The maximum amount of pulling (tensile) stress that a material can withstand before breaking or failing.

Elastic Modulus (or Young's Modulus): A measure of a material's stiffness. It defines the relationship between stress (force applied) and strain (how much it

deforms) in the elastic region, indicating how much a material will stretch or deform under a load and still return to its original shape. The higher the value, more force it requires to cause it to deform by a certain amount.

Thermal Conductivity: A material's intrinsic ability to conduct or transfer heat. A material with high thermal conductivity (like aluminium) transfers heat quickly, while a material with low thermal conductivity (like foam) acts as an insulator.

Propulsion Concepts

Electrical Power: The rate at which electrical energy is transferred or consumed in an electric circuit, typically measured in Watts (W). It is the product of voltage (V) and current (A), representing the energy drawn from the source (like a battery) by a component (like a motor).

Mechanical Power: The rate at which work is done by a mechanical system, measured in Watts (W). In this context, it is the rotational power delivered by the motor's shaft to the propeller, calculated as the product of torque and angular velocity (RPM).

Torque: The rotational equivalent of linear force. In the context of this UAV's propulsion system, it represents the twisting force generated by the electric motor's shaft. This force is what spins the propeller against air resistance, allowing it to generate thrust. A motor with higher torque can spin a larger diameter propeller at lower RPMs, which is often more efficient for endurance cruise. It is typically measured in Newton-meters (N·m).

Kv Rating: A specification for a brushless electric motor indicating its no-load rotational speed (in RPM) for every one volt of electricity applied. A lower Kv motor generally produces more torque per amp, while a higher Kv motor produces more speed.

ESC (Electronic Speed Controller): A device that controls the speed and direction of an electric motor by modulating the power delivered from the battery. It also often handles tasks like telemetry and battery protection.

C-Rating: A specification for a battery that defines its maximum safe continuous discharge rate, expressed as a multiple of its capacity (C). For example, a 10C, 2000mAh battery can safely discharge at $10 \times 2000\text{mA} = 20\text{A}$.

Static Thrust: The amount of thrust (force) produced by a propulsion system (like a propeller or jet engine) when the vehicle is held stationary (zero velocity) relative to the surrounding fluid.

SoC (State of Charge): The level of charge of a battery relative to its total capacity, expressed as a percentage. 100% SoC means fully charged, while 0% SoC means fully discharged.

Formulas

Electrical Power (W): $P = V \times I$

Mechanical Power (W): $P = (\tau \times \text{RPM} \times 2\pi) / 60$

Reynolds Number: $Re = (\rho \times v \times L) / \mu$

Lift coefficient: $C_L = (2 \times L) / (\rho \times v^2 \times A)$

Drag coefficient: $C_D = (2 \times D) / (\rho \times v^2 \times A)$

τ : Torque (N·m)

V: Voltage = 37V

I: Current (A)

τ : Torque (N·m)

ρ : Fluid density = 0.909 kg/m³ (at 3000m)

v: Fluid velocity (in m/s)

L: Characteristic length = 0.31 m (mean aerodynamic chord)

μ : Dynamic viscosity of the fluid = $\sim 1.694 \times 10^{-5}$ N·s/m² (at 3000m)

L: Lift Force (N)

D: Drag Force (N)

A: Reference Area = 1.03 m² (wing area)